

Monel 400 | Corrosion-Resistant & High-Strength Nickel-Copper Alloy

Introduction:

Monel 400 (UNS N04400) is a solid-solution nickel-copper alloy with excellent corrosion resistance in marine, chemical, and high-temperature environments. Known for high strength, toughness, and good weldability, it performs reliably in oxidizing and reducing conditions, making it ideal for pipes, fittings, valves, and industrial components.

Chemical Composition:

Element	Composition (%)
Nickel	63.0 min
Copper	27.0 - 33.0
Carbon	0.25 max
Manganese	1.5 max
Iron	2.0 max
Sulfur	0.01 max
Silicon	0.5 max
Aluminum	2.30 - 3.15
Titanium	0.35 - 0.85

Physical & Mechanical Properties:

- **Density:** 8.80 g/cm³
- **Melting Range:** 1300-1350 °C
- **Tensile Strength:** 55-100 ksi
- **Yield Strength:** 25-50 ksi
- **Elongation:** 70-120%
- **Hardness:** 60-100 HRB

Available Products:

- Seamless & Welded Pipes (ERW & EFW)
- Tubes (Capillary, U-Bend)
- Sheets, Plates, Coils & Strips
- Round, Flat, Hex & Square Bars
- Pipe Fittings, Flanges & Fasteners
- Custom Fabricated Components

Applications:

- Marine & Shipbuilding (propeller shafts, heat exchangers, valves)
- Chemical & Petrochemical Processing (tanks, vessels, pumps)
- Oil & Gas (downhole tools, pump shafts, safety valves)
- Power Generation (boiler systems, condensers, superheaters)
- Aerospace & Defense (fuel tanks, hydraulic components)
- Water Treatment & Desalination (brine heaters, evaporators)
- Electronics & Instrumentation (connectors, terminals, housings)

Advantages:

- Exceptional corrosion resistance in seawater, acids, and alkalis
- High strength and toughness across wide temperature ranges
- Good weldability and formability
- Non-magnetic and high conductivity
- Long service life with minimal maintenance

Why Choose Moksh Tubes & Fittings LLP:

- ISO-certified supplier of Monel 400 products
- Specialized in premium nickel alloys with global reach
- Quick delivery, custom fabrication, and certified quality
- Trusted by marine, chemical, oil & gas, and aerospace industries